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Education

Purdue University

Bachelor of Science in Computer Science

Concentrations: Software Engineering, Systems Programming

Minor: Business Management

- College of Science Deans List & Semester Honors: 2023, 2024, 2025
- Relevant Coursework: *Data Structures and Algorithms (C++)*, *OOP*, *Discrete Mathematics*, *Calculus*, *Linear Algebra*, *Software Engineering*, *Computer Architecture (Asm Arm64)*, *Programming in C*, *Systems Programming*, *Operating Systems*.

West Lafayette, IN

May 2027

GPA: 3.85/4.0

Skills

Languages C, C++, C#.Net, Python, JavaScript (TS, React), HTML/CSS (TailwindCSS), Java, Swift, Lua, VimScript, R, Bash, AppleScript, Assembly (x86, Arm64)

Tools L^AT_EX, Vi/Vim/Neovim, Zsh, Unity, Airflow, Linux, Debian, PostgreSQL, AArch64, Ubuntu, Unix, Git

Experience

C Programming Teaching Assistant @ Purdue University

November 2024 – Present

- Provided academic support to 750+ students (CS 24000), leading lab sessions, holding weekly office hours, and clarifying concepts related to the C programming language
- Developed a homework assignment from scratch to reinforce data structure fundamentals using LaTeX.
- Implemented multiple midterm and final practice exams into the class curriculum to enhance students' exam readiness and practical skills.

Software Engineering Intern @ Qservus

December 2022 – March 2023

- Conducted implementation of Qservus Lite for two months, a product oriented in quality of service, using AngularJS.
- Streamlined the deployment of the international version of Qservus' main website.
- Developed a mobile version (iOS) of a customer survey application using Swift.

Projects

SlayerHIL — C (protobuf-c), C++ (Physics engine, UDP, SPI), Raspberry Pi, RTOS

August 2025 - Present

Embedded Hardware-in-the-Loop Drone (UAV) Testbed

- Engineered a reproducible drone HIL platform to test off-aircraft flight software through simulating dynamics and automating runs/results through a Raspberry Pi-based orchestrator.
- Developed a C++ flight-physics engine for an RTOS, injecting deterministic telemetry and aggregating results over SPI using Protocol Buffers for controlled iterative aircraft testing.

BoilerMate — TypeScript (React), PostgreSQL, Supabase, Prisma ORM, NestJS, AWS, AI

August 2025 – January 2026

Roommate Matching Solution

- Designed a roommate discovery website to streamline Purdue University's off-campus housing search, leveraging a NestJS/PostgreSQL backend to deliver a social media implementation powered by LLMs (Google AI Studio).
- Led coordination and onboarding of team members through Agile/SCRUM principles, shipping with over 5,000+ impressions and partnering with Purdue faculty stakeholders for expanding outreach.

BLShell — C++, C, UNIX, Flex, Bison

September – November 2025

Shell Command Line Tool for UNIX-based OS

- Built a Unix-based shell in C++ that tokenizes/parses user input (Flex/Bison) and executes commands via POSIX system calls, supporting arguments and standard I/O redirection.
- Engineered support for advanced features including interrupts, zombie process handling, builtin functions, environment variables, regex, and history, with an optimized memory allocator for running processes made in C.

Elite Edge — JavaScript (React), TailwindCSS, Python, PostgreSQL, Apache Airflow

May - September 2025

College Basketball Weekly Insights & Analytics

- Architected a Dockerized backend to scrape weekly KenPom data (2013–2025), store it in PostgreSQL, and schedule a Python (scikit-learn) model running with Apache Airflow for delivering NCAA advanced analytics.
- Delivered a mobile-responsive React and TailwindCSS website, allowing users to compare excellence trends and similarity scores for Top 20 NCAA teams across 11 seasons through 2025.

macOS Neovim — Bash, AppleScript, Lua

February – April 2025

Desktop Application

- Developed an app enabling users to open files directly in a Neovim instance or the configured default file explorer, automatically handling shell sessions and directory management, using Bash and AppleScript.
- Implemented customizable application builds supporting multiple terminal emulators (Terminal.app, iTerm), including automated naming, icon integration, and streamlined macOS code signing.